

Transformer Representation Geometry in Non-Ergodic Sequence Models

A PyTorch transformer interpretability report on residual-stream geometry, latent belief-state decoding, and non-ergodic sequence modeling

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Executive Summary

This report presents an end-to-end PyTorch experiment studying how a small decoder-only transformer represents latent structure in a non-ergodic next-token prediction task. The model was trained on sequences generated from a mixture of three hidden Markov generative processes. Each sequence came entirely from one process, but the model was never given the process label. The analysis asks whether the residual stream encodes the true Bayesian belief state of the active process and whether it separately encodes which process generated the sequence.

The main result is a strong asymmetry. The ground-truth belief state is almost perfectly linearly decodable from residual-stream activations, with R^2 remaining above 0.989 across context positions. By contrast, process identity is only partially linearly decodable, reaching roughly 60% accuracy by the final context position compared with a 33% chance baseline. This suggests that the transformer learns a compact, shared, low-dimensional internal geometry that prioritizes local predictive belief state over clean global process separation.

This makes the experiment a useful toy model for AI safety and mechanistic interpretability. It provides a setting where the true latent variables are known exactly, allowing transformer activations to be compared directly against the mathematical belief geometry of the data-generating process.

Motivation and Relevance to AI Systems

Real language-model data is not generated by one stationary source. Text comes from many hidden regimes, including authors, domains, tasks, topics, styles, registers, and user intents. Each regime can be internally coherent, while the overall training corpus is a mixture of regimes. In this sense, non-ergodicity is not an edge case for language. It is a central feature of real data.

This project builds a minimal controlled version of that setting. Each training sequence is generated from one of three hidden stochastic processes, but the transformer must infer the active process only from the observed tokens. To predict well, the model must solve two nested inference problems: it must infer the local belief state within the active process, and it must infer the global process identity.

This structure is relevant to large language models because in-context learning can be viewed as latent process identification. When a model reads a prompt, it must infer the task, domain, style, or instruction regime that generated the context before making good local predictions. The experiment here makes that inference problem mathematically explicit and measurable.

The experiment also connects to computational neuroscience. In predictive coding and latent-state models of behavior, higher-level variables often represent context or regime, while lower-level variables track fast-changing state within that regime. The non-ergodic sequence setup provides a clean mathematical analogue of that same nested inference problem.

Experimental Setup

Data-generating processes

I used a mixture of $K=3$ Mess3 processes with distinct parameters: P1 with $\alpha=0.95$ and $x=0.02$, P2 with $\alpha=0.90$ and $x=0.05$, and P3 with $\alpha=0.85$ and $x=0.12$. Each process has a differently shaped belief-state attractor within the 2D probability triangle. A single Mess3 process trained in isolation is expected to produce a clean 2D belief geometry in the residual stream. The goal here was to study what happens when multiple such processes are mixed in a non-ergodic way.

Every training sequence of length 16 was generated entirely from one of P1, P2, or P3. The vocabulary consisted of symbols $\{0, 1, 2\}$ plus a beginning-of-sequence token. The model was never given the process label, so the only way to identify the generating process was to accumulate statistical evidence across the token sequence.

A key implementation detail was that the process parameters needed to produce meaningful sequential structure. Initial choices led to entropy rates close to $\log(3)$, making the token stream nearly uniform and difficult to learn from. Switching to high-persistence and low-noise parameters created genuine sequential dependencies, allowing the model to learn and enabling meaningful representation analysis.

Transformer model and training

The model was a small decoder-only transformer implemented directly in PyTorch. It used two layers, a model dimension of 64, two attention heads, and an MLP width of 256, for approximately 101K parameters. The transformer processed sequences of up to 17 tokens, consisting of a beginning-of-sequence token followed by 16 observed tokens.

The training objective was next-token cross-entropy prediction. The model was trained for 8,000 gradient steps with the Adam optimizer, a learning rate of $5e-4$, and a batch size of 256. The model converged to a cross-entropy loss of approximately 0.74, well below the uniform-token baseline $\log(3)$, which is approximately 1.099. This confirmed that the transformer learned predictive sequential structure rather than only matching marginal token frequencies.

Analysis pipeline

After training, I extracted residual-stream activations across layers and context positions. I computed exact ground-truth Bayesian belief states from the known Mess3 transition matrices, then compared those belief states to the model activations. The main analyses were PCA-based effective dimensionality, linear decoding of belief state, linear decoding of process identity, and geometric visualization of residual-stream activations.

Training and Representation Results

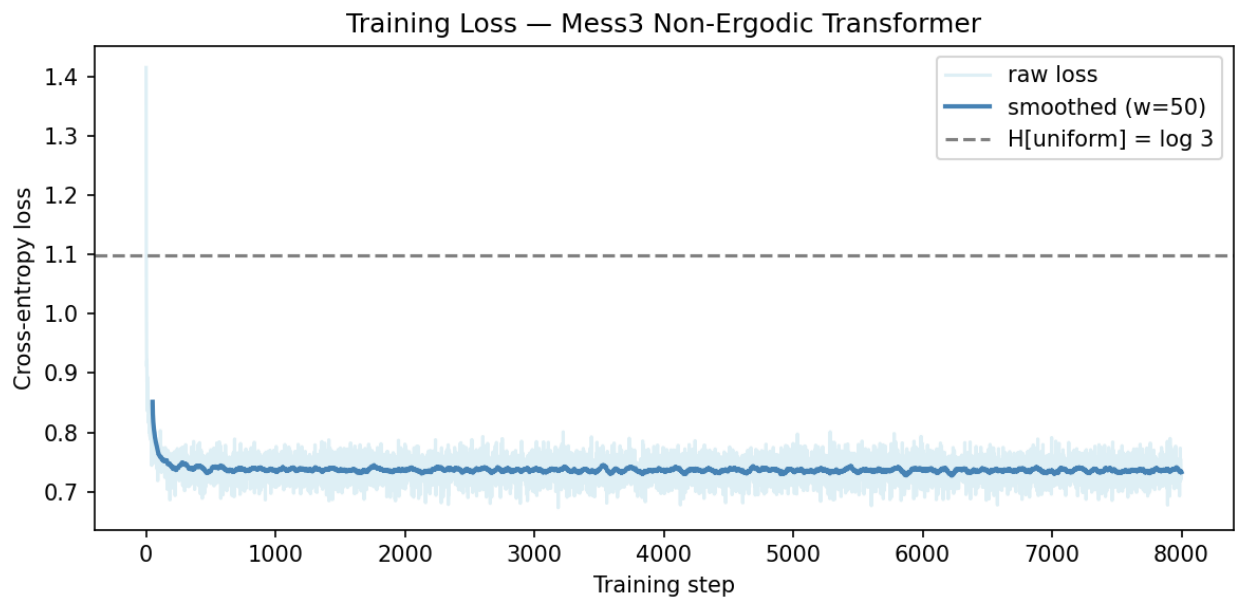


Figure 1. Training loss. Cross-entropy loss over 8,000 training steps for the transformer trained on the non-ergodic sequence dataset. The loss briefly spikes near initialization, then drops rapidly within the first few hundred steps and converges around 0.74. The dashed reference line at $H[\text{uniform}] = \log 3 \approx 1.099$ represents a trivial uniform predictor over the three-symbol alphabet. The converged loss substantially below this baseline confirms that the model learned genuine predictive structure in the non-ergodic data.

The clean convergence and stable final loss provide the foundation for the geometric analyses. The residual-stream structure analyzed below is meaningful because the model successfully learned the sequence task. A model that failed to infer structure would not be expected to show interpretable belief-state geometry.

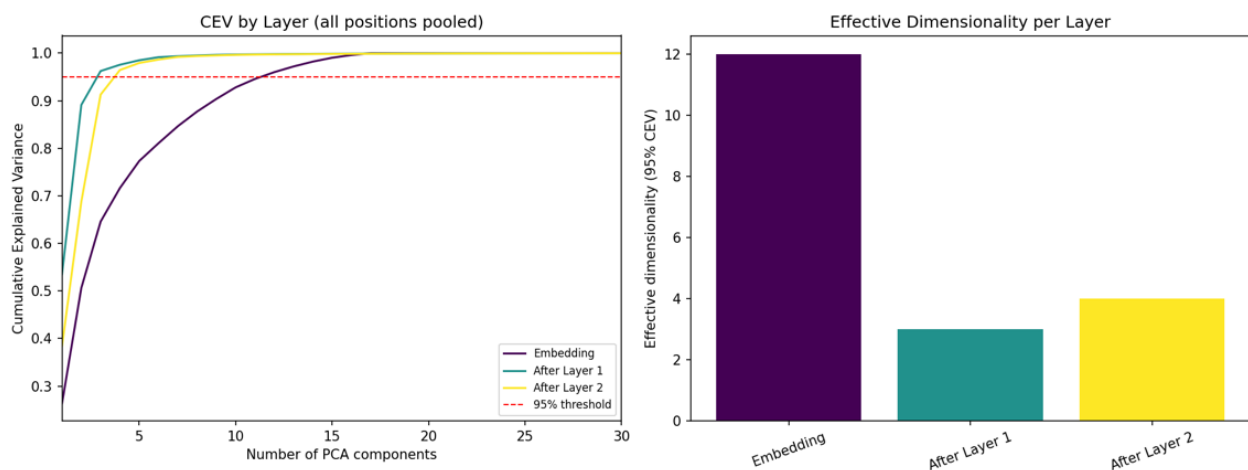


Figure 2. Cumulative explained variance by layer and effective dimensionality. The raw embedding layer requires about 12 PCA components to cross the 95% explained-variance threshold. After the first transformer layer, this collapses to roughly 3 components, and after the second layer it stabilizes around 4 components. This sharp compression indicates that the transformer reorganizes high-dimensional token and position embeddings into a compact low-dimensional internal representation.

The final dimensionality of roughly 4D is below the 6D upper bound expected from three fully separate 2D belief geometries. This suggests that the model shares representational dimensions across processes rather than assigning each process a cleanly orthogonal subspace.

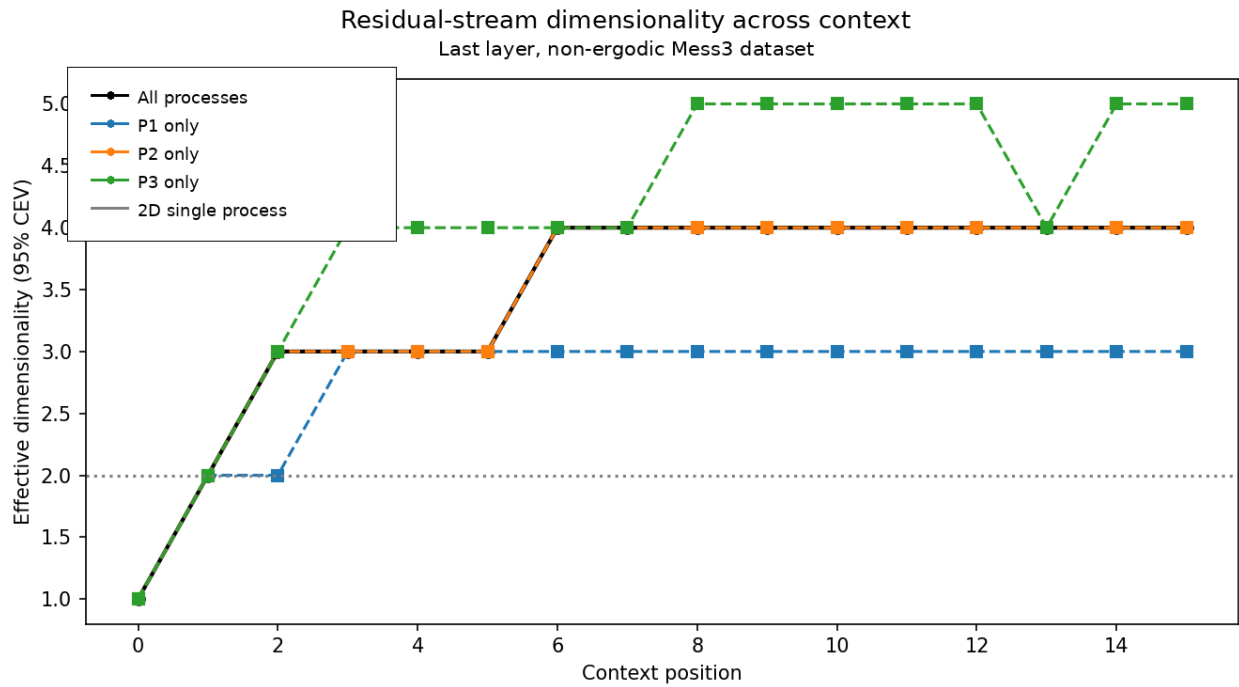


Figure 3. Effective dimensionality across context position. The gray dotted line marks the 2D reference for a single 3-state belief geometry. The effective dimensionality grows with context as the transformer accumulates evidence about both hidden state and process identity. Individual processes exceed the 2D single-process reference, indicating that the model carries additional structure beyond within-process belief state. P3 shows more erratic and higher-dimensional behavior, consistent with a more complex belief attractor that is harder to identify from finite context.

The context-position analysis addresses the core non-ergodic inference problem. At the beginning of a sequence, the model has little information about the active process. As context accumulates, the residual stream expands to represent both local belief state and uncertainty over the generating process.

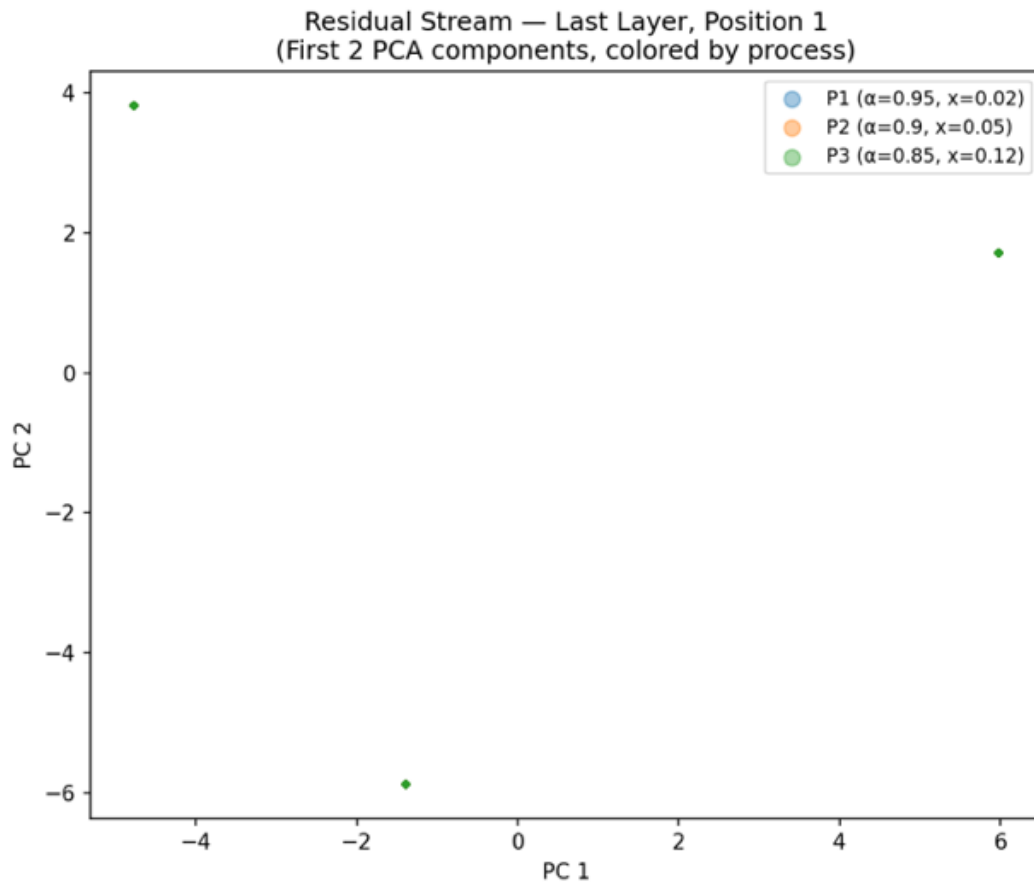


Figure 4. Residual-stream PCA at early context. At position 1, after a single observed token, the representation contains three isolated points corresponding to token identity. The processes overlap almost completely, showing that one token carries little information about which generative process is active.

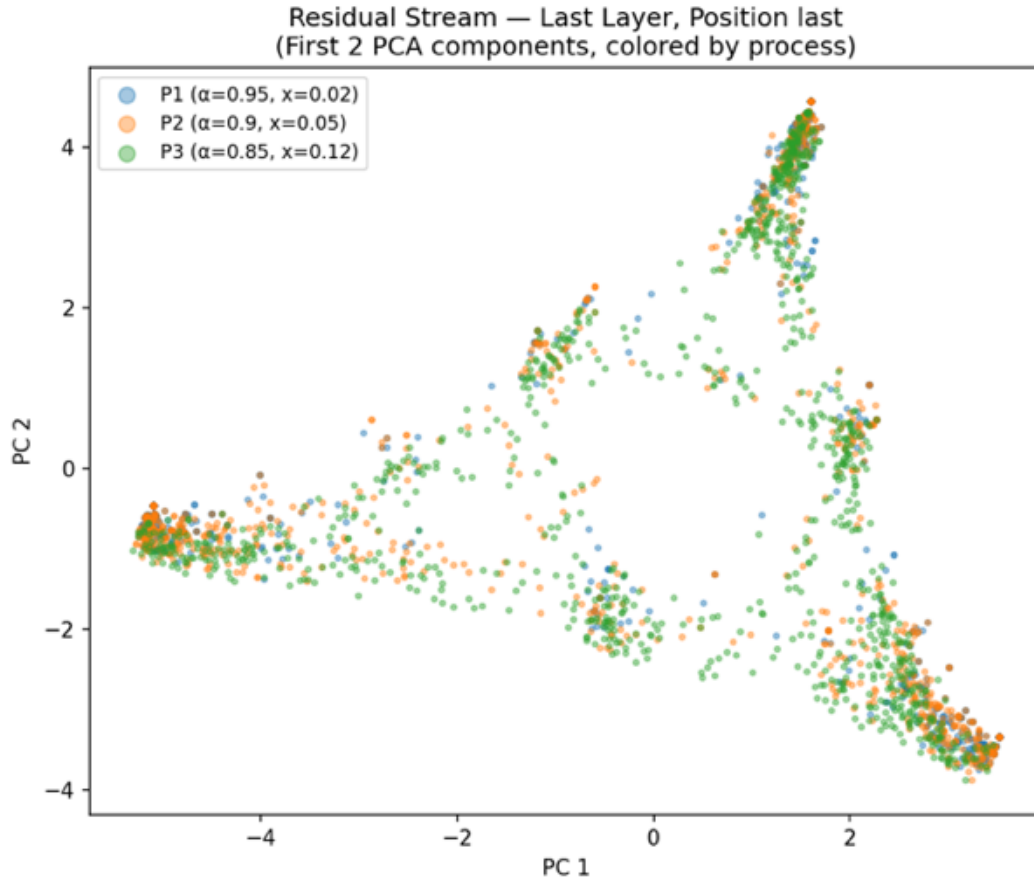


Figure 5. Residual-stream PCA at late context. By the final context position, the residual stream develops a rich branching geometry. The representation shows partial process separation, but not clean process-specific clusters. This supports a shared non-orthogonal geometry rather than separate orthogonal subspaces for each process.

The early-to-late contrast shows how the residual stream changes with context. Early activations encode local token identity. Late activations form a structured branching geometry that reflects accumulated evidence about latent state and process identity.

Mathematical Prediction for the Geometry

Before analyzing the trained model, I derived a prediction for the geometry of the residual stream. Each Mess3 process has three hidden states, so its belief vector $\mathbf{b} = (b_0, b_1, b_2)$, with the entries summing to 1, lives in a 2-dimensional probability triangle. A lossless linear representation of a single process therefore requires 2 dimensions.

In the non-ergodic mixture of three processes, the full joint belief state at context position l is a distribution over process index k in $\{1, 2, 3\}$ and hidden state s in $\{0, 1, 2\}$. This joint posterior can be written as $p(k, s \mid \mathbf{x}_{1:l}) = p(k \mid \mathbf{x}_{1:l}) p(s \mid k, \mathbf{x}_{1:l})$. It is a normalized 9-dimensional vector and therefore formally lives in an 8-dimensional probability space.

If the process identity were perfectly inferred, all probability mass would concentrate on one process block and the representation would collapse back to a 2D within-process belief vector. But because the process label is never directly observed, the model must maintain uncertainty over both process identity and hidden state throughout the context window.

The expected effective dimensionality lies between two useful bounds. A compact shared representation could use 2 dimensions for process identity and 2 dimensions for within-process hidden state, giving a lower bound of about 4 dimensions. A fully separated representation could allocate a distinct 2D belief

subspace to each of the three processes, giving an upper bound of about 6 dimensions.

My prediction was that the learned geometry would fall closer to the lower bound. Because the three processes share a vocabulary and have related structure, I expected the model to reuse a shared belief-state geometry across processes, with only partial additional dimensions devoted to process identity.

Candidate geometries

- A fully factored geometry would encode process identity and within-process belief in cleanly separated subspaces. Both belief state and process identity would be nearly perfectly linearly decodable.
- A fully joint geometry would encode the entire joint distribution over process and hidden state without clean factorization. This would be faithful but higher-dimensional.
- A partially shared geometry would reuse a common belief-state subspace across processes while only partially separating process identity. This predicts excellent belief-state decoding but weaker process-identity decoding.

The empirical results most closely match the partially shared geometry.

Ground-Truth Belief Geometry and Linear Decoding

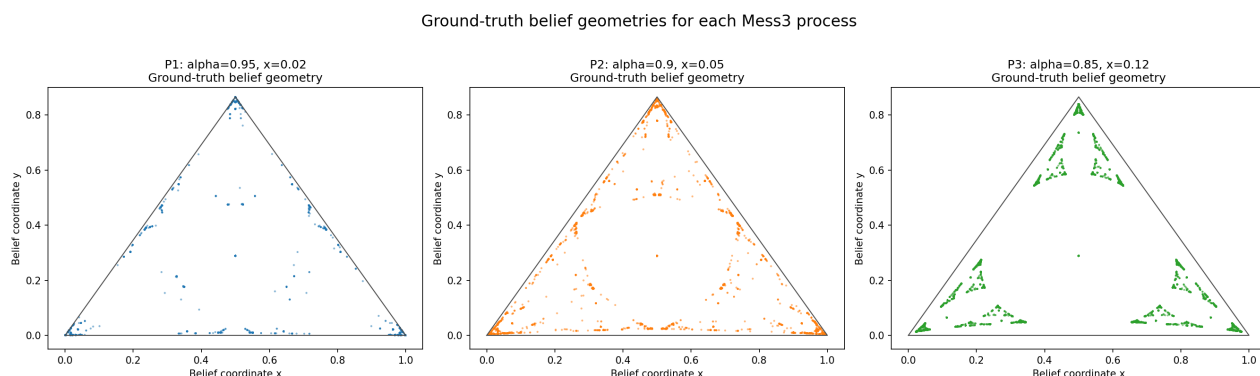


Figure 6. Ground-truth belief geometries. Belief-state trajectories for the three Mess3 processes plotted in barycentric belief-coordinate space. P1 is sparse and concentrated near probability-triangle vertices and edges, P2 forms a clearer triangular edge structure, and P3 shows a more complex multi-scale triangular attractor. These qualitatively different geometries are the targets that the transformer must internalize.

For the residual-stream analysis, I computed the exact ground-truth belief state at every context position for each validation sequence using the Mess3 transition matrices. I then tested whether these belief states could be decoded from the residual stream using linear maps.

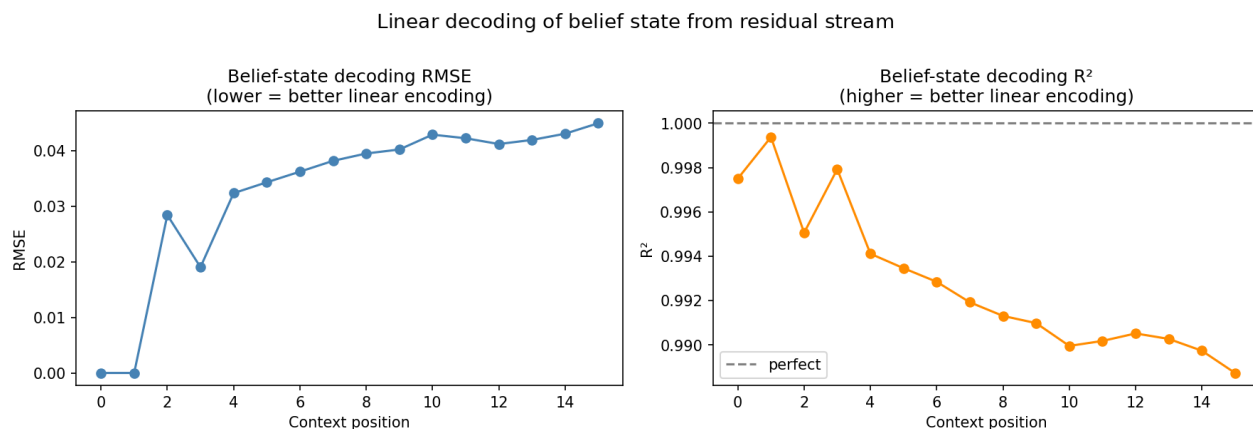


Figure 7. Linear decodability of belief state from the residual stream. The true belief state is linearly recoverable with very high fidelity across the entire context window. R^2 peaks near 0.999 and remains above 0.989 throughout. RMSE remains small, increasing gradually at later positions as the geometry becomes richer. This confirms that the transformer linearly encodes the belief geometry of the generative process.

This is the strongest result of the project. The model was trained only on next-token prediction, yet its residual stream contains a representation that is nearly linearly equivalent to the true Bayesian belief state.

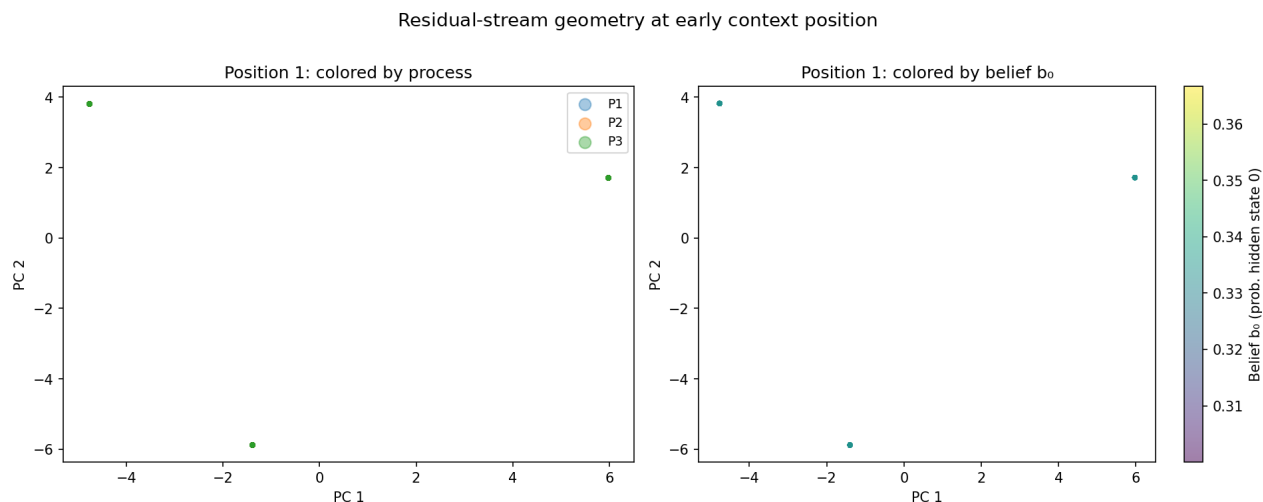


Figure 8. Early-context residual stream colored by process identity and belief coordinate. At position 1, all processes collapse to the same three token-driven points. The belief coordinate b_0 remains close to one-third, confirming that the posterior is near-uniform after a single token. The residual stream differentiates token identity but not process identity at this stage.

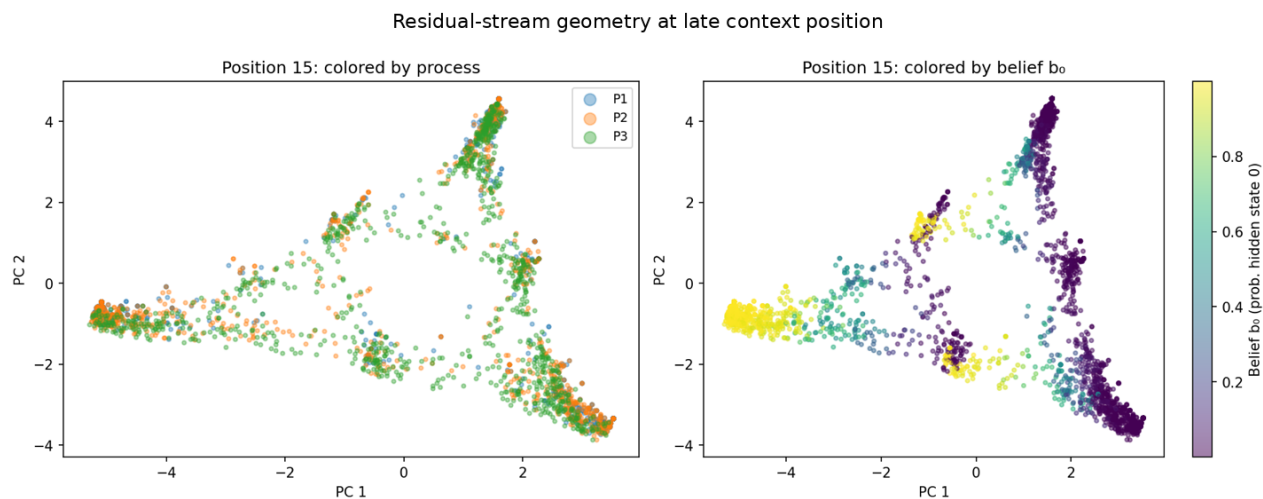


Figure 9. Late-context residual stream colored by process identity and belief coordinate. By position 15, the residual stream forms a Y-shaped branching cloud. The b_0 coordinate varies smoothly across the geometry, with high values in one dense region and low values near the branch tips. This smooth gradient indicates that nearby points in activation space correspond to nearby belief states.

The late-context geometry shows that the transformer has learned a continuous representation organized by belief coordinates. The process labels are partially separated but remain mixed, which supports the conclusion that the model uses a shared non-orthogonal activation space rather than cleanly separated per-process subspaces.

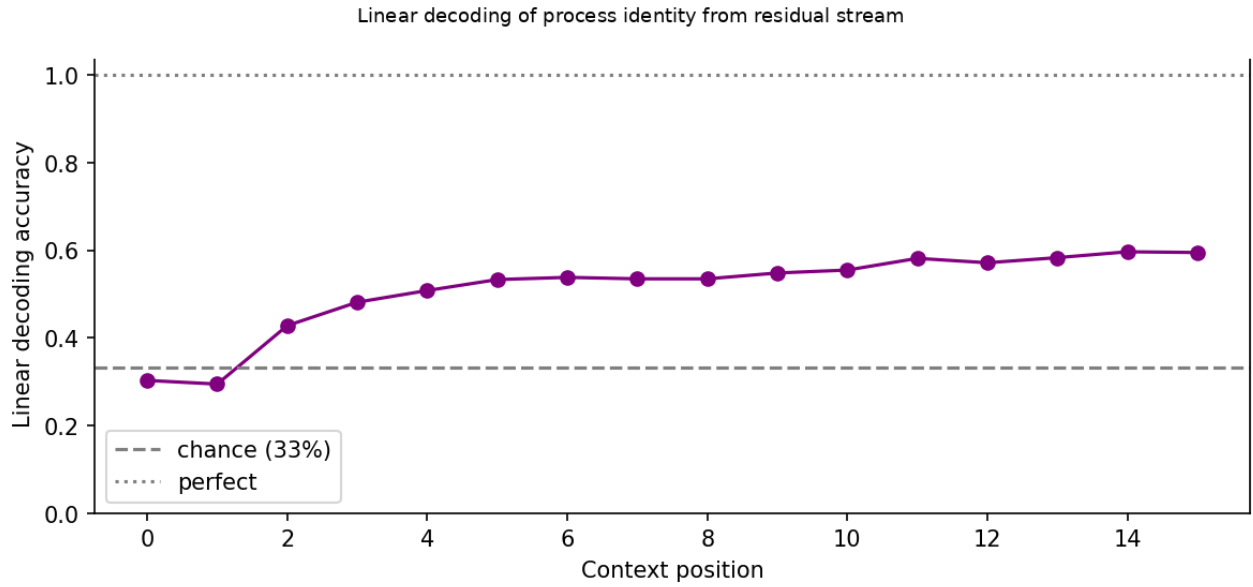


Figure 10. Linear decoding of process identity. Process identity decoding starts near chance and gradually improves with context, reaching approximately 60% by the final position. This is well above the 33% chance baseline but far below perfect classification. The model therefore accumulates information about the active process, but process identity is not represented as cleanly as belief state.

The contrast between Figures 7 and 10 is the central asymmetry. Belief state is almost perfectly linearly decodable, while process identity is only partially decodable. The transformer appears to prioritize local predictive structure over explicit global source separation.

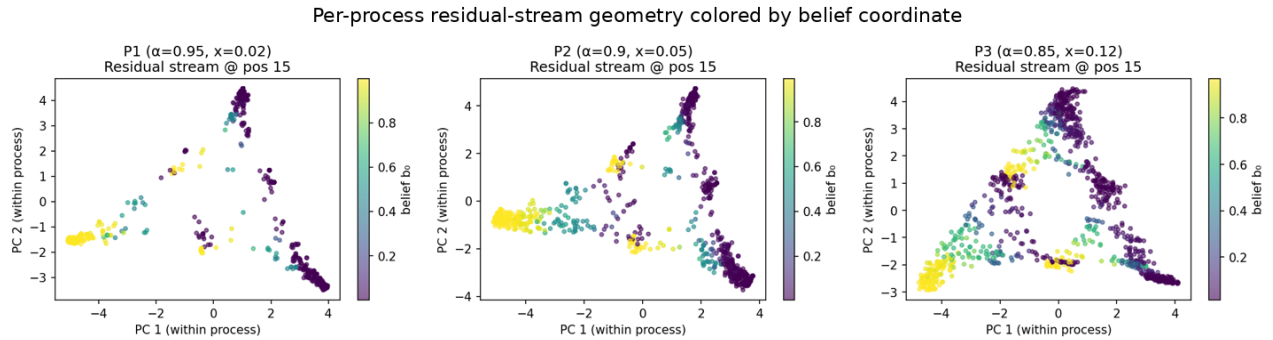


Figure 11. Per-process residual-stream geometry at late context. Each panel shows within-process PCA components at position 15, colored by belief coordinate b_0 . All three processes show branching multi-armed structure with smooth belief-coordinate gradients. The density and spread differ across processes, reflecting their different parameters, but the same basic belief-organized structure appears in each case.

Interpretation

The trained transformer develops a low-dimensional residual-stream geometry that closely mirrors the ground-truth belief geometries of the component processes. At early context, the representation is dominated by token identity and contains almost no process-specific structure. At late context, the residual stream forms a rich branching geometry in which belief coordinates vary smoothly across activation space.

The result is not consistent with fully orthogonal process-specific subspaces. Instead, the three processes share a common activation geometry. Process identity becomes partially recoverable only as context accumulates. This is exactly what should happen in a non-ergodic setting where process identity must be inferred gradually rather than being directly observed.

The finding has implications for AI safety and interpretability. It shows that a transformer trained with next-token prediction can internally represent belief-like variables, even when those variables are never supervised. It also shows that prediction quality does not imply clean representation of global context. A model can track local predictive state extremely well while only partially representing which broader regime it is in.

In larger language models, the analogous latent variables may be task, topic, user intent, instruction regime, or domain. A model may predict locally coherent next tokens while remaining uncertain or entangled about the global regime. This provides one possible mechanistic lens on prompt sensitivity and inconsistent behavior across similar prompts.

Additional Analyses and Extensions

Scaling the number of processes

A natural extension is to train on 2, 5, or 10 processes and track how effective dimensionality changes. I expect early-context dimensionality to grow with the number of possible processes, because there are more alternatives to disambiguate. At late context, dimensionality may collapse toward a lower-dimensional belief geometry if the model successfully identifies the active process. If dimensionality remains high, that would indicate that the model continues to represent unresolved process uncertainty.

Varying process similarity

Another extension is to vary how similar the processes are. When two processes have nearly identical parameters, they should be harder to distinguish from short contexts. I would derive an analytical distinguishability measure as a function of parameter distance and compare it to empirical process-decoding accuracy over context length. This would provide a quantitative test of whether the model performs process identification in a statistically efficient way.

Mid-sequence process switching

A more challenging experiment is to generate sequences that switch processes halfway through, for example first eight tokens from one process and then eight from another. This would test whether the residual stream updates dynamically when the latent regime changes, or whether the model becomes stuck on its initial process inference. This is directly relevant to how language models handle topic, style, or task shifts within a single document or conversation.

Subspace-overlap analysis

The PCA visualizations suggest that the processes share an activation subspace rather than occupying orthogonal subspaces. A stronger quantitative analysis would compute normalized subspace overlap between per-process activation subspaces. Values near zero would indicate orthogonal representations, while values near one would indicate a fully shared subspace. This would turn the qualitative shared-geometry conclusion into a precise number.

Bayesian optimality

Another useful analysis would compare the linear decoding RMSE against theoretical limits for belief estimation from finite sequences. This would test whether the residual stream approximates near-optimal Bayesian filtering, or whether the non-ergodic mixture imposes a measurable representational cost.

Connection to latent-state inference in neuroscience

This experiment connects to latent-state models of decision-making in neuroscience, where animals can switch between discrete behavioral strategies and the goal is to infer those hidden states from observations. A future analysis could fit a regime-switching latent-state model directly to transformer

residual-stream trajectories and test whether the model resolves process identity through discrete internal transitions or through smooth interpolation. This would connect mechanistic interpretability methods with computational neuroscience tools for studying latent state dynamics.

Skills Demonstrated

- PyTorch implementation of a decoder-only transformer from scratch
- Causal self-attention, residual connections, layer normalization, and next-token prediction
- Synthetic non-ergodic sequence generation from hidden Markov processes
- Exact Bayesian belief-state computation from known transition matrices
- Residual-stream activation extraction and geometry analysis
- PCA, cumulative explained variance, and effective dimensionality analysis
- Linear probing of latent belief state and process identity
- AI safety and mechanistic-interpretability framing using controlled toy models
- Research communication connecting mathematical predictions, model training, and empirical results

Conclusion

This project shows that a small PyTorch transformer trained on a controlled non-ergodic next-token prediction task learns a compact internal representation of latent belief state. The residual stream encodes the true Bayesian belief state with very high linear decodability, while process identity is represented more weakly and remains partially entangled with belief geometry.

The results support a partially shared representation: the model reuses a common belief-state geometry across processes rather than allocating fully separate orthogonal subspaces. This provides a clean example of how next-token prediction can induce interpretable latent variables inside transformer activations, while also showing that not all relevant latent structure is represented equally clearly.

As a small AI safety-relevant interpretability experiment, this report demonstrates how controlled synthetic tasks can be used to study internal representations, latent-state inference, and residual-stream geometry in transformer models.